

Hobby Box EV — Stakeholder Report

To: hobby-box collector deciding whether to buy a sealed box · From: EV Calculator project · Prices as of: 2026-08-28 (Blowout street price / Topps SRP)

1. Bottom line

At today's prices, only one of the six sealed hobby boxes is a positive-EV buy — 2025-26 Topps NBA Hoops Hobby (≈ \$260, expected card value ≈ \$290). The two most-hyped premium boxes, Chrome Update Series and Cosmic Chrome Hobby (both ≈ \$1,100), return roughly \$0.20 of expected card value per \$1 paid — you are buying the chase-card lottery ticket, not the expected contents.

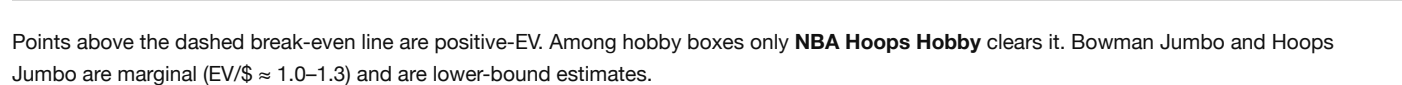
The buy/pass ranking is robust: halving or doubling our card-value estimates does not flip a single verdict, and how we handle missing pull-odds does not move the model.

2. Problem & method

Decision: should the collector pay price P for a specific sealed box? **Answer:** compare P to **EV** — the probability-weighted resale value of the cards inside.

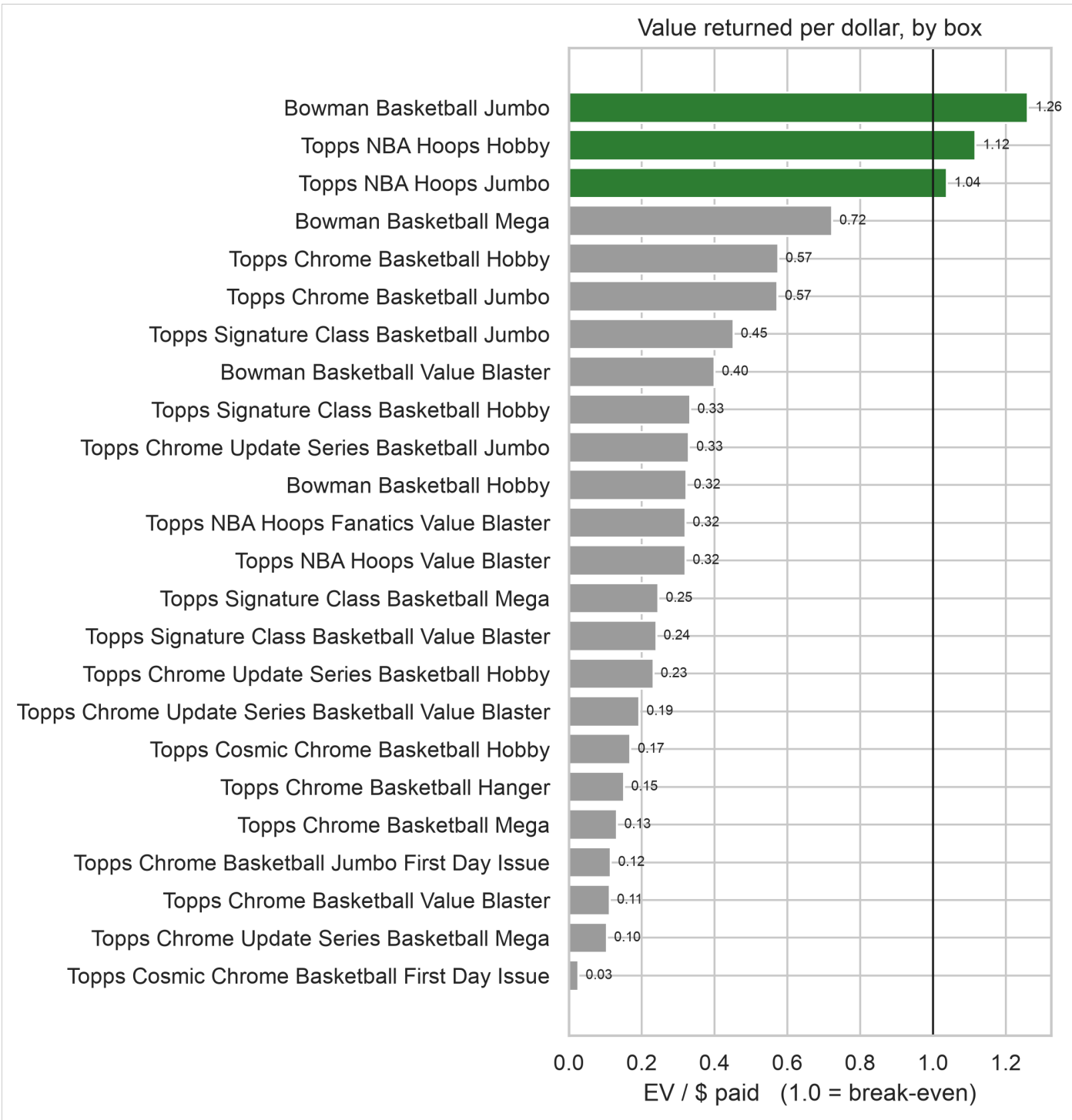
$$\begin{aligned} \text{EV_box} &= \sum \text{over card tiers } [(\text{packs_per_box} / \text{pull_odds}) \times \text{tier_value}] + \text{base cards} \times \$0.20 \\ \text{EV} / \$ &= \text{EV_box} / \text{price} \quad (> 1 = \text{good bet over many boxes}; < 1 = \text{pay-to-play}) \end{aligned}$$

- **Pull odds** from Topps / checklistinsider odds sheets (data/raw/card_tiers.csv).
- **Tier values** are a placeholder ladder (rough_estimate_v0), calibrated to the six real eBay sold comps collected so far (Alter Egos / Minions).
- Hobby boxes use the full per-tier ladder; other formats use a floor estimate (autographs + SSP inserts + base only). Implementation: src/ev.py ; full table data/processed/ev_report.csv .



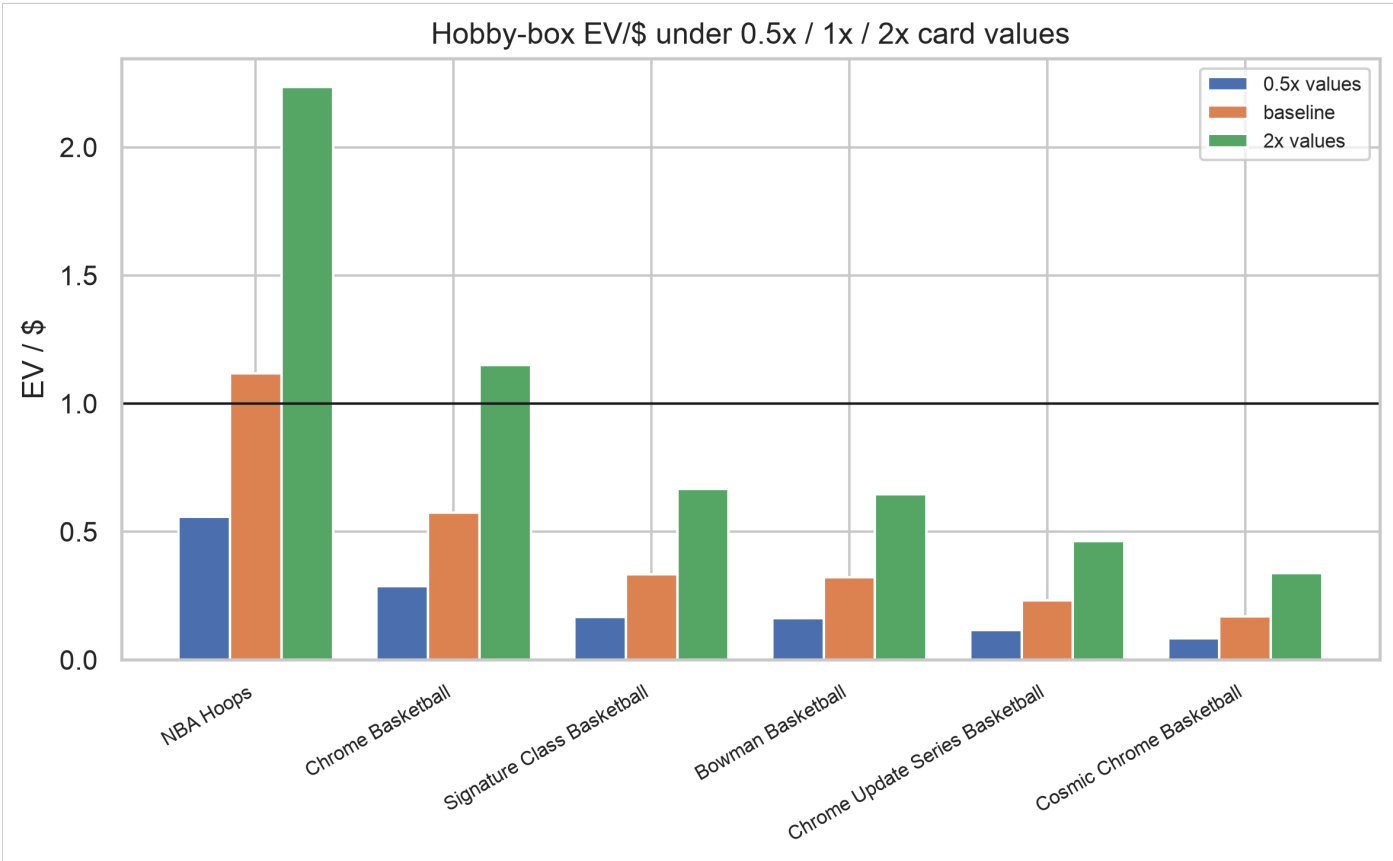
Points above the dashed break-even line are positive-EV. Among hobby boxes only **NBA Hoops Hobby** clears it. Bowman Jumbo and Hoops Jumbo are marginal ($EV/\$ \approx 1.0\text{--}1.3$) and are lower-bound estimates.

3.2 Value returned per dollar



box	price	EV	EV/\$	verdict
NBA Hoops Hobby	\$260	\$290	1.12	buy
Chrome Basketball Hobby	\$380	\$219	0.58	pass (EV)
Signature Class Hobby	\$550	\$183	0.33	pass (EV)
Bowman Basketball Hobby	\$500	\$162	0.32	pass (EV)
Chrome Update Series Hobby	\$1,100	\$255	0.23	pass (EV)
Cosmic Chrome Hobby	\$1,100	\$185	0.17	pass (EV)
Chrome Jumbo <i>First Day Issue</i>	\$3,500	\$401	0.11	avoid

3.3 Sensitivity — does the verdict survive our biggest assumption?



box	EV/\$ @0.5x	EV/\$ baseline	EV/\$ @2x
NBA Hoops	0.56	1.12	2.23
Chrome Basketball	0.29	0.57	1.15
Signature Class	0.17	0.33	0.67
Bowman	0.16	0.32	0.65
Chrome Update Series	0.12	0.23	0.46
Cosmic Chrome	0.08	0.17	0.34

No verdict flips. Every premium box stays "pass" even at 2x; NBA Hoops stays "buy" even at 0.5x. Chrome Basketball Hobby only becomes marginal (1.15) at 2x.

4. Assumptions & Risks

assumption	baseline	alternate tested	effect on the decision
Card values = rough_estimate_v0 ladder (6 real comps)	EV as reported	×0.5 / ×2	ranking robust ; absolute EV is indicative, not precise
Missing pull-odds → column-mean impute	slope 0.669	median / drop (scenario_results.csv)	slope 0.669 / 0.670 / 0.669 — no change
EV is a mean; box outcomes are right-skewed	report EV	report median / P(beat price)	most boxes return below EV — only buy when EV/\$ is comfortably > 1
Non-Hobby EV omits the parallel rainbow	autos + SSP + base	full tier ladder (not transcribed)	non-Hobby EV/\$ shown are lower bounds
Prices are one snapshot (2026-08-28)	Blowout / SRP	live market feed	hot-box prices move weekly — re-run before acting
Selling costs	ignored	~13% eBay/PayPal fees	subtract before treating any EV/\$ ≈ 1 as profit

5. Decision implications & next steps

For the collector

- **Buy:** NBA Hoops Hobby at/under ~\$260 — the only sealed box that clears break-even on expected contents.
- **Pass (as an EV play):** Chrome Update, Cosmic Chrome, Signature Class, Bowman, Chrome Basketball — all return < \$0.60 per \$1. Buy only for the rip or a specific chase, eyes open.
- **Avoid on EV:** retail Blaster / Mega / Hanger, and the Chrome Jumbo First Day Issue (\$3,500, EV/\$ ≈ 0.11).

For the tool

- Replace the placeholder ladder with real eBay sold-comp medians — start with autographs and low-numbered parallels.
- Transcribe per-format tier odds so non-Hobby EV is a real estimate, not a floor.
- Weekly price refresh so EV/\$ tracks the live market.
- Report $P(\text{box beats price})$ and the median outcome next to EV — the mean hides the right-skew.

Reproduce: `python src/ev.py` · `notebooks/homework12_...submission.ipynb` · Stage 11 sensitivity in `notebooks/homework11_...ipynb`.